



University
of Glasgow

April/May Exam Session 2025
Duration: 1 hour 30 minutes
Timed exam fixed start time

DEGREES OF MSc, MSci, MEng, BEng, BSc, MA and MA (Social Sciences)

TEXT AS DATA FOR MSC/MSCI COMPSCI 5096/5106

(Answer all 3 questions)

This examination paper is an open book, online assessment and is worth a total of 60 marks.

1. **Question on TF-IDF and Set Similarity. (Total marks: 20)**

Consider the following vocabulary constructed from an unknown language. Inverse Document Frequency (IDF) values are provided and rounded to the nearest integer for ease of arithmetic.

Token ID	Token	IDF
0	A	2
1	B	4
2	C	1
3	D	3
4	E	6
5	F	4
6	G	1
7	H	3
8	I	3
9	J	5

- (a) Construct a dense TF-IDF bag-of-words representation for the following document. Use the IDF values provided above and linear TF scaling. Tokenize on whitespace.

DocX: G C G A H A G

[4]

Example Answer: Representation: [4, 0, 1, 0, 0, 0, 3, 3, 0, 0]

- (b) Consider the following TF-IDF bag-of-words representation for DocY, which used the same procedure as in (a). Reconstruct the original text for DocY. Specify any ambiguities encountered during reconstruction.

DocY: [6, 8, 0, 6, 6, 0, 2, 3, 3, 10]

[4]

Example Answer: The document contains: A A A B B D D E G G H I J J. However, the order of the words is unknown, so they may have appeared in any order.

- (c) Compute the Overlap Coefficient and Jaccard Similarity between DocX and DocY. Show your work.

[4]

Example Answer:

Set representation of DocX: A, C, G, H

Set representation of DocY: A, B, D, E, G, H, I, J.

Cardinality of DocX: 4
Cardinality of DocY: 8
Cardinality of DocX intersect DocY: 3
Cardinality of DocX union DocY: 9
Overlap = $3/4 = 0.75$
Jaccard = $3/9 = 1/3 = 0.33333...$

(d) A colleague proposes a new set similarity measure:

$$\text{sim}(A, B) = \frac{|A - B|}{|A \cap B|}$$

Note that $A - B$ represents the set difference operation (in other words, the elements of set A that are not in set B). Compute $\text{sim}(\text{DocX}, \text{DocY})$. Show your work.

[2]

Example Answer:

(Building on work from part (c))

Cardinality of DocX - DocY: 1

$\text{sim}(X, Y) = 1/3 = 0.33333...$

(e) Describe **three** potential problems with the proposed similarity measure from (d).

[6]

Example Answer: *Note: there are many problems. Here are three examples:*

- It measures distance, not similarity (larger value = less similar)
- Its asymmetric ($\text{sim}(A, B) \neq \text{sim}(B, A)$)
- Could get a divide by 0 error when A and B are disjoint

2. **Question on Text Classification. (Total marks: 20)**

- (a) You are building a multi-class text classification system to categorize recipes for an existing recipe website using the text of the recipe. It has four classes: comfort food, easy-to-make, Italian and pasta.

To build your labelled dataset, you search the current website using the existing search functionality and the class names (e.g. searching for "comfort food"). You take the first five recipes retrieved for each class as your labelled data.

Discuss **four** problems that this approach may have.

[8]

Example Answer: *Note: there are many problems. Here are four examples:*

- This is a multi-label problem and not a multi-class problem. Some documents (e.g. an Italian pasta recipe) may be labelled by more than one of the given classes which a multi-class approach would not be able to cover.
- Five recipes for each label is far too few. It would likely not be enough to train and evaluate a system.
- Good evaluation data needs to be annotated by a person. A purely automated approach for generating labels would likely give lower quality annotations which would cause poorer performance for the system.
- The classifier would do no better than the existing search functionality. By using the search functionality to construct and label the dataset, the maximum achievable performance is what is already achieved by the search functionality.

- (b) You realise that samples with the "comfort food" label have been triplicated (repeated three times) by accident in the dataset. This error occurred before any data splitting occurred. What effect would this have on the training and evaluation? Give **four** points.

[4]

Example Answer:

- May cause class imbalance
- Class imbalance may cause problems during training with an over-focus on "comfort food"
- There may be copies of the same sample in the training and evaluation (validation/test) data, causing test set leakage which invalidates any performance results

- The created class balance will not match the real dataset so some metrics may not be realistic

(c) A separate classifier is trained and evaluated for identifying the sentiment of user comments on the reviews. Consider the statistics below. Identify **eight** anomalies in the dataset, the presented metrics and the overall training/evaluation procedure.

[8]

		Predicted	
		Pos	Neg
Actual	Pos	2	2
	Neg	8	3

Table 1: Confusion matrix for training set

Sample Count	15
Accuracy	0.5
Perplexity	19.2
Macro-Precision	0.2
Macro-Recall	0.5
Macro-F1	0.6

Table 2: Metrics for training set

		Predicted		
		Pos	Neg	Neutral
Actual	Pos	50	0	0
	Neg	0	10	0
	Neutral	0	0	5

Table 3: Confusion matrix for validation set

Sample Count	65
Accuracy	1.00
Perplexity	10.8
Macro-Precision	1.00
Macro-Recall	1.00
Macro-F1	1.00

Table 4: Metrics for validation set

Example Answer:

- Perplexity is not a metric used for classification problems
- The training set has two classes, but the validation set has three classes
- The accuracy for the training set doesn't match the confusion matrix
- The training set has Neg as the largest class, but the validation set has Pos as the largest class.
- Against best-practices, the validation set has substantially more samples than the training set.
- The performance on the training set is poor, but it is very good for the validation set

- The F1 score for the training set is wrong as it should be between the precision and recall values.
- Macro precision/recall are not typically used for binary problems, and the values are actually for precision/recall

3. **Question on Sentiment Analysis. (Total marks: 20)**

Consider a situation where you are working for a company, BeStep, that is looking to monitor its brand online. You've been hired to work on a project that tracks the sentiment (positive, negative, neutral, or mixed sentiment) towards the brand on social media.

(a) You are considering three possible approaches:

1. BERT-base, a pre-trained language model that is fine-tuned on your labelled sentiment data.
2. A logistic regression model that uses TF-IDF features and is fine-tuned on your labelled sentiment data.
3. LLAMA-70B, a Large Language Model that is prompted for sentiment.

Rank the three approaches in terms of their computational cost of deployment in this application, from least expensive to most expensive. Justify your answer.

[2]

Example Answer: logistic regression < BERT < LLAMA.

Logistic regression is a lightweight linear classifier over simple features, while BERT and LLAMA are larger neural networks. BERT and LLAMA use a similar neural architecture (transformer), but LLAMA is much larger (uses more layers, etc) than BERT.

(b) Describe a procedure for determining which of the three solutions is most effective in this setting. Clearly state any data you need to determine effectiveness in this setting.

[4]

Example Answer: You need samples of social media posts that mention your brand and are labeled with one of the four sentiment classes. The data should be split into training, validation (dev), and test sets. You should then train and tune each of the methods using the training and validation data, and compare the results of the best version of each model with the test set to determine the most effective one. You should evaluate using classification measures like Precision, Recall, and F1.

(c) BeStep wants to expand internationally next year and, therefore, wants to ensure it is ready to monitor social media brand sentiment in languages other than English. Give one problem you would encounter when applying a logistic regression model in another language. Give a different limitation that you would encounter when applying a BERT-base classifier in another language. Suggest a solution that addresses both these problems.

[4]

Example Answer:

Logistic regression relies on tokens observed during training, with unseen tokens being replaced by a placeholder. Most tokens in different languages will be seen as unknown, substantially reducing the effectiveness of your classifier.

BERT-base splits words into sub-tokens, so they will not be marked as unknown. However, since Word-Piece tokens are trained on a corpus, if the language was not included in the model's pretraining data, so most words from other languages will be split into meaningless subwords.

Note: There are several possible solutions. Here is one example:

- You could collect and train new models on the new language.

- (d) You consider a fourth approach, a BERT-based model that is fine-tuned on open benchmark data for sentiment classification. However, when you run it on your data, you find that it classifies every sample in your validation data that mentions the brand name as "positive" sentiment. You check the false positives (examples below) and confirm that their gold labels are correct.

- I disliked my experience with bestep.
- @Bestep I placed an order three weeks ago, and still haven't gotten it!
- I'm not happy with my purchase BeStep. I want a refund.

The problem is being caused by **sub-word tokenisation**. Describe the problem and provide one suggestion on how to address it.

[4]

Example Answer: The classifier probably tokenizes bestep as best ##ep (or similar). Since the model has an input that includes the word "best", it considers the sample as positive sentiment, since this is typically an indication of positive sentiment.

Note: there are many ways to resolve this issue. Here is one example:

- You could replace the name of the brand "BeStep" with a placeholder like "BRAND". This will address the problem because the placeholder won't tokenize to any sentiment-related words.

- (e) BeStep wants a more fine-grained understanding of the sentiment towards each of their 18 products. Describe an approach you would use to accomplish this goal and how it would fit into the existing sentiment classification system.

[6]

Example Answer:

Note: there are many possible approaches. Here is one example:

- You could perform multi-label classification over sentiment labels for each product+sentiment pair, including one for "no product"+sentiment. You would need training and evaluation data on a per-product basis to assess this system. This system would replace the existing system entirely, since the "no product"+sentiment class would cover the existing case.